Aman Kumar

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EXPERIENCE

NativeLink **Remote (California, US)**
Software Engineer *June 2025 – Present*

- Integrated **Google Cloud Storage** as a CAS backend, extending support for large-scale caching and remote execution.
- Upgraded and maintained **Helm Charts** for Kubernetes (GKE), ensuring reliable and production-ready deployments.
- Developed and validated **Bazel build configurations** for enterprise customers, accelerating onboarding and evaluation.
- Enhanced distributed build performance by working across the **backend**, **Kubernetes**, and **Bazel** ecosystem.

Fortanix Inc. **Bengaluru, India**
Software Developer II *Feb 2025 – June 2025*

- Designed a **distributed caching layer**, improving scan performance and reducing service latency.
- Built a **Post-Quantum Cryptography (PQC) Readiness Dashboard** to analyze cryptographic usage across products.
- Built the **CBOM Export Too**l to identify and track cryptographic libraries/components across code, filesystems, and containers, ensuring compliance with secure key management standards.
- Contributed to secure key management and **enclave-based workflows** (Intel SGX, Azure, AWS) for production-grade Data Security Manager deployments.
- Increased test coverage from **32%** to **66%**, improving reliability through comprehensive unit/integration tests.

Software Developer I *Mar 2024 – Feb 2025*

- Optimized scan workflows by refining streaming patterns and enhancing the in-house **Azure SDK**.
- Implemented core features for Key-Insight including **On-Premise Scan**, **Federated Authentication**, and **Policy Center APIs**.
- Expanded unit/integration test coverage for scan flows and CRUD APIs.
- Developed high-performance backend services in **Rust** and **Python** with a focus on secure, scalable cryptographic workflows and distributed systems.

Software Development Intern *Sep 2023 – Mar 2024*

- Extended the in-house **Azure SDK** with hierarchy and IAM support, enabling scalable cloud resource scanning.
- Contributed to CRUD API design and integration for Key-Insight.
- Implemented and secured cryptographic key management workflows in Data Security Manager, with hands-on experience in **Intel SGX enclaves**, **cloud HSMs** (Azure, AWS), and secure storage.

iRage Capitals **Mumbai, India**
Quantitative Analyst Intern *Dec 2022 – May 2023*

- Built a **real-time risk management system** with alert dashboards to monitor trading activity and threshold breaches.
- Analyzed derivatives data (PnL, IV Gap) to identify trading patterns and improve performance.
- Automated trading workflows using **Shell Scripting**, **Linux**, and **Python** (data pipelines, analytics, and report generation).

GNOME Foundation (*Google Summer of Code*) **Remote (Berlin, Germany)**
Software Development Intern *May 2022 – Sep 2022*

- Reduced application startup time by **32%** by optimizing database migration logic.
- Contributed automation scripts in **Python** to streamline testing and development workflows.
- Added **multi-user functionality** by migrating schemas to Tracker DB.
- Optimized performance of a Linux application stack using **Rust, GTK**, and **TrackerDB**.

PROJECTS

Order Matching Engine | *GitHub* **May 2023**

- Implemented a NASDAQ ITCH Limit Order Book in **Rust**, designed for low-latency, high-throughput trading.
- Achieved processing speeds of **11.4M messages/sec** with latency as low as **87 ns/tick**.
- Optimized data structures and concurrency primitives to ensure predictable performance under heavy load.

Distributed Cache System | *GitHub* **Jan 2025**

- Developed a fault-tolerant, horizontally scalable distributed cache in **Rust** with **consistent hashing** and multiple eviction policies (**LRU, LFU**).
- Implemented replication, leader/follower coordination, and **failure recovery mechanisms**, demonstrating knowledge of distributed consensus trade-offs.
- Designed **thread-safe concurrent operations** and optimized memory access patterns for low-latency read/write performance.

EDUCATION

Indian Institute of Information Technology, Nagpur **Nagpur, India**
Bachelor of Technology - Computer Science and Engineering *Jul 2019 - Jun 2023*

TECHNICAL SKILLS

Languages: Rust, C/C++, Python, SQL, JavaScript/TypeScript

Backend & Systems: Distributed Systems, Concurrency/Parallelism, Caching, Cryptography, REST/gRPC APIs, Tokio, Actix, Flask, Django, Cryptographic Systems, SGX, Confidential Computing

DevOps & Cloud: Bazel, Kubernetes (GKE), Helm, Docker, CI/CD, Google Cloud (GCS, GKE), Azure, AWS

Tools & Platforms: Linux, Git, Shell Scripting, GTK